GAN-Based Sub-Instance Augmentation for Open-Pit Mine Change Detection in Remote Sensing Images

Zilin Xie, Jinbao Jiang, Deshuai Yuan, Kangning Li, and Ziwei Liu

***Abstract*— Remote sensing (RS) change detection (CD) for open-pit mines plays a critical role in both mineral development and environmental conservation. The performance of supervised deep-learning-based CD for open-pit mines is often limited by the amount of available data, necessitating data augmentation. Current CD data augmentation methods mainly focus on image-, region-, and instance-level transformations. However, mining area changes occur at a finer sub-instance level, leading to suboptimal performance of existing methods. Therefore, this study proposes a novel data augmentation method named generative adversarial network (GAN)-based sub-instance augmentation (GSIA). This method enables the generation of realistic and diverse CD samples using unchanged data from mining areas to address the issue of data scarcity in open-pit mine CD. GSIA comprises three steps. In the first two steps, GSIA achieves sub-instance-level transformations by sequentially applying GAN-based local editing to the labels and images of the mining areas. In the third step, GSIA constructs a synthetic CD dataset by randomly combining the bitemporal data. The effectiveness of GSIA was evaluated by comparing it with 14 other data augmentation methods on five CD models, and GSIA outperformed all of them. In addition, training solely on synthetic data generated by GSIA achieved an overall accuracy of 97.64% and an $F1$-score of 78.65%, which were comparable to training with all available real data. Furthermore, GSIA can address the insufficient correlation between the training and test sets in domain adaptation. GSIA plays a significant guiding role in the data augmentation for open-pit mine CD. Our code is available at https://github.com/ZilinCode/GSIA.**

***Index Terms*— Change detection (CD), data augmentation, generative adversarial networks (GANs), open-pit mine, sub-instance-level transformation.**

I. Introduction

MINERALS are among the most important natural resources worldwide. Mineral resource mining, particularly in open-pit mines, causes severe ecological damage [1], [2]. Rational and orderly mineral development is crucial to protect ecosystems and mineral resources. Therefore, change detection (CD) in surface mining areas is necessary to effectively supervise the mining development process. Currently, methods based on remote sensing (RS) images are widely applied in open-pit mine CD [3], [4], [5], [6], and the availability of high-resolution satellite data has opened new avenues for fine-grained CD [7]. However, owing to the great complexity of high-resolution RS image information, the extraction of mining area change information based on high-resolution RS is still dominated by manual visual interpretations by professionals [8].

In recent years, supervised deep-learning techniques have gained increasing attention for open-pit mine CD because of their powerful ability to extract image information from large amounts of labeled data [8], [9], [10], [11]. However, collecting effective data for open-pit mine CD is challenging because of the limited number of change areas. Furthermore, annotating CD datasets is time-consuming and labor-intensive, and it may even be impossible to obtain sufficient data despite searching all research areas. For example, the open-pit mine CD dataset used by Du et al. [8] contained only 45 sets of data. In addition, the change areas in the annual open-pit mine CD were finer and more difficult to annotate manually. However, without sufficient training data, supervised deep-learning-based CD models may lack the necessary generalization ability to be applied to complex feature information and imaging conditions in RS.

Data augmentation provides a solution for supervised deep learning with limited data, and it can be categorized into three scales of transformations: image level, region level, and instance level. Methods based on image-level transformations, such as flipping, rotation, color shifting, and noise addition [12], [13], [14], [15], [16], have been widely used. Subsequently, several methods with region-level transformations were proposed, including Cutout [17], CutMix [18], and Mosaic [19]. Recently, several instance-level transformation methods were introduced, including ObjectAug [20] and background swapping (BS) [21]. Most of these methods rely on expanding the existing datasets and may not effectively address the limitations of small or biased samples. Such limitations could hinder the generalization and robustness of the CD models.

To address this issue, several methods have recently been proposed to augment the training dataset by synthesizing new CD data from RS images of other regions. These methods

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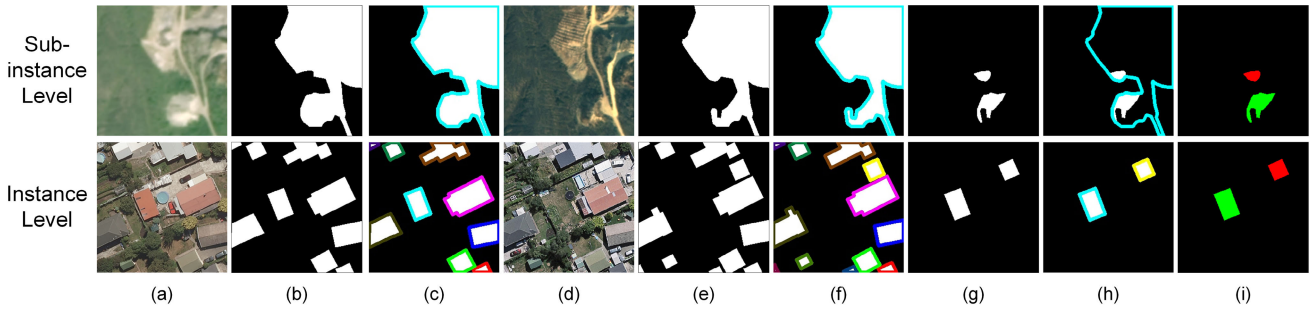


Fig. 1. (Top layer) Comparison between the open-pit mine CD data and (bottom layer) WHU building CD data [25]. By extracting boundaries from the feature labels, various instances can be delineated [22]. (a) Image at the first time point ($T1$). (b) Feature label at $T1$. (c) Feature label with instance boundaries at $T1$ (different colors represent different instances). (d) Image at the second time point ($T2$). (e) Feature label at $T2$. (f) Feature label with instance boundaries at $T2$. (g) Change label. (h) Change label with instance boundaries. (i) Semantic change label (red denotes feature increase and green denotes feature decrease).

include instance patch insertion in image methods [22], [23], generative adversarial network (GAN)-based image generation methods [13], and GAN-based local image editing methods [24]. Some of these methods, such as that proposed by Luo et al. [13], perform global transformations at the image level with a high degree of randomness, which requires laborious manual labeling of the data. Other methods [22], [23], [24] do not require manual annotation. However, these methods have been developed for instance-level transformations, such as the appearance and disappearance of objects and their overall attribute changes. These methods may not apply to scenarios where fine-scale changes occur within local regions of instance, such as in mining areas. As illustrated in Fig. 1, each building change area corresponds to the building, representing the overall appearance or disappearance of the detection target, i.e., instance-level change. By contrast, each change in the mining area only accounts for a fraction of the entire area, representing local expansion or contraction of the detection target. We refer to the finer scale than the instance level as the sub-instance level.

GAN-based local image editing with mask guidance enables region-level transformations and has been used for object shape editing [26], [27]. Rui et al. [24] employed instance labels as masks to apply their model to instance-level transformations. When smaller masks are used, with the maximum active ranges defined by the instance labels and their surrounding areas, the model can be used for sub-instance-level transformations. Based on this, we propose a novel data augmentation method, called GAN-based sub-instance augmentation (GSIA). GSIA comprises three steps. In the first two steps, GSIA achieves sub-instance-level transformations of labels and images sequentially, by applying improved GAN-based local image editing models. Such transformations can generate areas of change from the unchanged areas. In the third step, GSIA constructs a synthetic CD dataset by randomly combining the bitemporal data. Our approach effectively addresses the problem of data scarcity in open-pit mine CD.

The contribution of this work can be summarized as follows.

1) We propose a novel data augmentation method called GSIA to address the challenge of data scarcity in CD for open-pit mines. Unlike typical instance-level methods, GSIA belongs to sub-instance-level transformations, allowing for modifications to partial regions of instances.
2) The performance of GSIA does not degrade with the reduction in the original CD data quantity but is enhanced instead because it can use data from other unchanged mining areas for augmentation. Moreover, training solely on synthetic data generated by GSIA can achieve recognition accuracy comparable to training with all available real data.
3) The GSIA can be used not only for data augmentation but also to address the issue of insufficient correlation between the training and test sets in domain adaptation.

The remainder of this article is organized as follows. Section II presents studies related to the proposed method. Section III introduces the detailed architecture of GSIA. Section IV describes the experimental data and settings and evaluates the image generation quality and data augmentation capability of GSIA. Section V analyzes the parameters of GSIA and discusses its feasibility for domain adaptation. Finally, Section VI concludes this article.

II. Related Work

A. Data Augmentation for CD

Data augmentation is a widely used data-level solution in deep learning to improve the model detection performance by increasing the size and value of the training set. Current data augmentation methods can be categorized into three groups based on the scale of transformation: image-, region-, and instance-level methods. However, our proposed GSIA, which enables transformations at a finer scale than the instance level, does not belong to any of these categories but falls under the sub-instance-level methods.

Image-level augmentation methods refer to transforming the entire image, including flipping, rotation, cropping, translation, color adjustment, and noise addition. Moreover, in the field of RS, setting a certain overlap rate or using random cropping can be considered interpolation-based data augmentation methods [8], [11]. In recent years, GAN-based data augmentation for image generation has gained increasing attention [13], [28], [29], [30]. Most of image generation methods belong to image-level methods.

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Region-level augmentation methods refer to applying specific transformations to certain regions of an image to increase diversity, including Cutout, CutMix, Mosaic, YOCO [31], and RandomMix [32]. In addition, GAN-based data augmentation for local image editing also belongs to region-level methods [33].

Instance-level augmentation is a specialized form of region-level augmentation. It involves specific transformations of instances or backgrounds based on feature labels, thereby aiding CD models in a better understanding of feature information. Instance-level augmentation methods include ObjectAug [20], BS [21], and so on. In addition, another advantage of instance-level methods is the ability to synthesize instance-level CD samples from unchanged RS images. For example, based on the characteristics of instance-level changes (appearance or disappearance of the target as a whole) in building CD, Yan et al. [22] and Chen et al. [23] have employed a method of pasting building instance patches onto background images for data augmentation. Rui et al. [24] also achieved instance-level change generation by setting the editing scope of GAN-based local image editing to building instances.

Our proposed GSIA belongs to the sub-instance-level augmentation methods, with a finer scale of transformations compared to the three existing augmentation methods mentioned above. Sub-instance-level methods extend instance-level methods by refining transformations, narrowing their scope from the entire instance to local regions of the instance. Similar to instance-level methods, the advantage of sub-instance-level methods is the ability to synthesize sub-instance-level CD samples from unchanged RS images, which is more applicable to open-pit mine CD.

B. GAN-Based Local Image Editing

Image editing primarily involves altering images through color and geometric interactions for tasks such as image deformation and blending [34]. To synthesize more realistic images, GANs have been utilized. GAN-based image editing methods can be categorized as local image editing or global image editing according to the editing area. Local image editing focuses on editing only a part of the image while maintaining other regions unchanged, whereas global image editing involves editing the entire image [35]. For example, local image editing techniques include deocclusion [26], [36], deshadowing [37], [38], object shape editing [26], [27], and facial local attribute editing [35], [39], whereas global image editing techniques include style transfer [40], [41], super-resolution [42], and facial global attribute editing [43], [44], [45].

Local image editing can ensure that other regions remain unchanged, which is advantageous for open-pit mine CD augmentation. First, the sub-instance-level transformation of the mining area data can be achieved through local image editing, which is not possible with either image generation or global image editing. Second, CD data have two characteristics: 1) differences between changed regions and similarities between nonchanged regions in bitemporal images and 2) consistency of images at the same time and possible dissimilarities in images at different times. Local image editing satisfies these requirements well. Finally, the local image editing is based on real RS images, which avoids the problem of complex feature generation in an image background. Therefore, GAN-based local image editing was adopted to augment the open-pit mine CD dataset.

III. Method

A. Overall Architecture

GSIA utilizes bitemporal semantic segmentation data from mining areas as the basis for a three-step process involving label generation, image generation, and dataset construction. To clearly illustrate the overall pipeline, the original data from the first time ($T1$) were temporarily retained while solely processing the data from the second time ($T2$), as shown in Fig. 2.

In the label generation phase, a label orientation mask (M_L) is used to mask the original label (L_O), enabling local orientation editing at a sub-instance level and generating a new label (L_G). Second, in the image generation phase, the original image (I_O) is masked by a mask containing mining area features (M_I), and L_G is used to guide local semantic editing at a sub-instance level, generating a new image (I_G). Finally, in the dataset construction phase, a synthetic open-pit mine CD dataset is constructed with the original I_O at $T1$, the generated I_G at $T2$, and a change label (L) obtained by subtracting the original L_O at $T1$ from the generated L_G at $T2$.

In the aforementioned process, we observed that a dataset can be generated by maintaining the $T2$ data unchanged and processing the $T1$ data, as well as by processing both $T1$ and $T2$ data and randomly combining the resulting data. Therefore, in the actual execution of the first and second steps, $T1$ and $T2$ data were mixed without considering their temporal correspondence. In the third step, the correspondence between the $T1$ and $T2$ data was reconsidered, and the original and generated data were randomly combined to construct a synthetic open-pit mine CD dataset. Two different combination strategies were employed during the third step: 1) combining the original data at $T1$ ($T2$) with the generated data at $T2$ ($T1$) and 2) combining the generated data at $T1$ and $T2$.

B. Label Generation

Zheng et al. [46] proposed a pluralistic image completion model, PICNet, which combines the strengths of variational autoencoder and GAN. This model can reconstruct specific regions of an image guided by a mask, serving as the foundation for constructing a label editing model. As shown in Fig. 3, PICNet consists of two branches: top and bottom. The bottom branch is the main pathway that uses the remaining parts of the image to obtain a prior distribution and random samples to complete the image, to generate diverse images. The top branch is an auxiliary pathway that uses the missing parts of the image to obtain a prior distribution and benefits from the prior distribution obtained from the bottom branch to assist the bottom branch. Using a parallel dual-pathway design, PICNet improves the diversity of bottom-branch image

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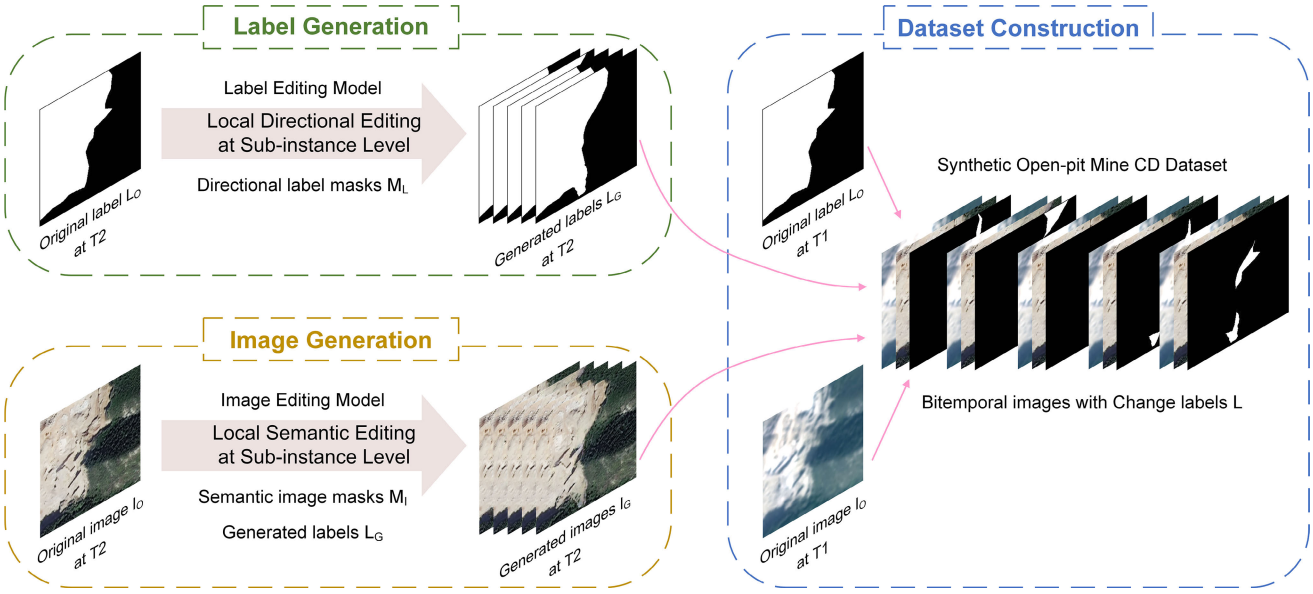


Fig. 2. Overall architecture of GSIA. It mainly consists of three steps: label generation, image generation, and dataset construction. Only the case of maintaining $T1$ data unchanged and processing $T2$ data is shown here.

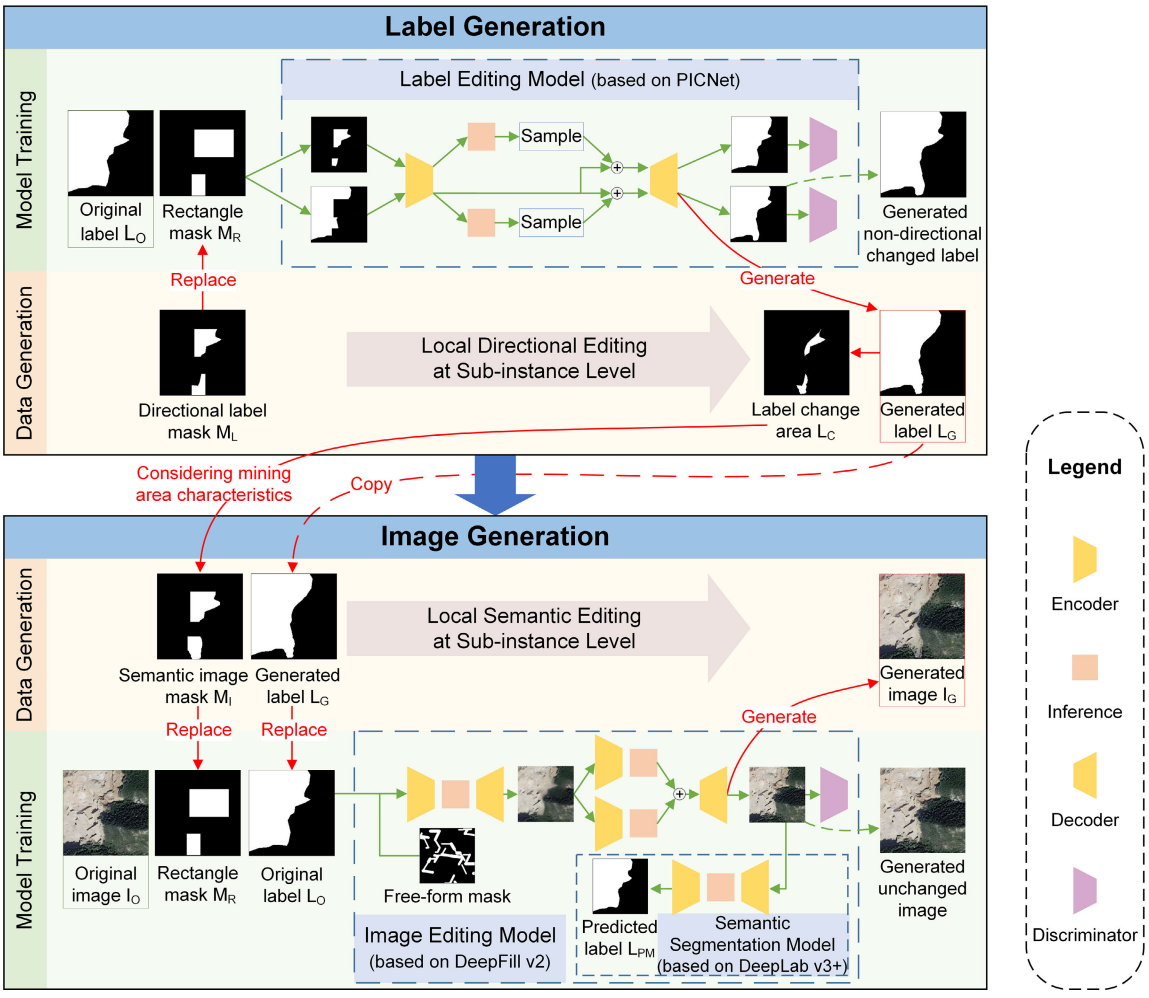


Fig. 3. Implementation details of label and image generation process in GSIA. In contrast to typical GAN-based models, GSIA uses different masks during model training and data generation (i.e., model testing) stages. During the data generation stage, specific masks are applied to guide the model in achieving local directional editing of labels and local semantic editing of images at sub-instance level, resulting in the synthesis of more realistic data.

generation, enabling greater diversity in the generated results with a similar mask.

As shown in Fig. 3, in contrast to the original execution process of PICNet, different masks are used for model training

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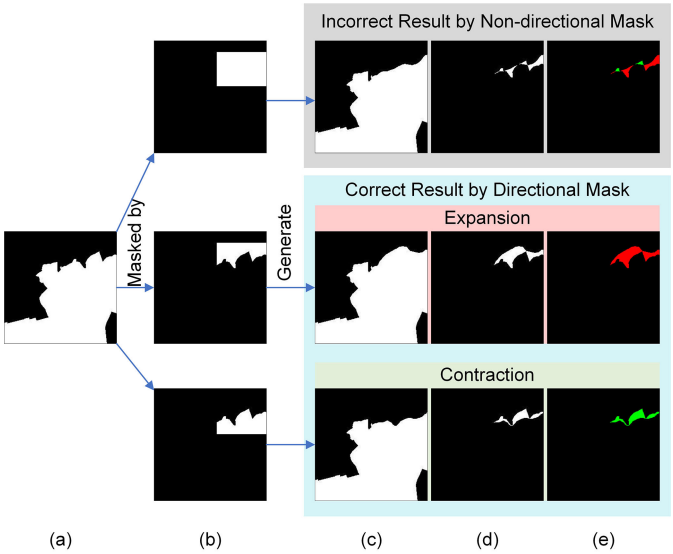


Fig. 4. (Top layer) Comparison between the effect of nondirectional and (middle and bottom layers) directional masks on the results during the data generation phase. (a) Original label L_O. (b) Different masks. (c) Generated label L_G. (d) Changed area L_C. (e) Semantic changed area (red denotes the expansion of the mining area and green denotes the contraction of the mining area).

and data generation (i.e., model testing) in the proposed approach.

During the PICNet training phase, the model must learn how to achieve pluralistic completion and generate reasonable mining area boundaries. Therefore, random rectangular masks (M_R) are generated at the mining area boundaries to serve as masked regions. M_R may contain multiple rectangular areas, serving two purposes: on the one hand, it helps to maintain a balanced dataset by appropriately increasing the proportion of changing regions relative to invariant regions [23]; on the other hand, instead of simply expanding the changed area, the process increases the frequency of changes, making the size of each generated changed area more realistic. Notably in our method, larger change areas are not always better, as detailed in Section V-A3.

During the data generation phase, a directed mask M_L is used instead of an undirected rectangular mask M_R. Each region in M_L covers only one side of the mining area boundary to guide directional changes, preventing PICNet from generating unreasonable results. Fig. 4 shows that if the mining area boundary is jagged, using only the undirected mask M_R leads to this phenomenon. Specifically, the mining area expansion and contraction appear in small areas within the generated change area (L_C) in an alternating manner. However, this phenomenon corresponds to the alternation of surface destruction and restoration in the mining area within a small area in reality, which is undesirable. Therefore, during the data generation phase, a directed mask M_L was used as the generated area to achieve directional editing and prevent this from occurring.

In addition, the output results of the PICNet may contain noise. Therefore, a postprocessing procedure was performed, as shown in Fig. 5.

The results in Fig. 5 demonstrate that when the mining area is small and mask M_L covers the entire area, PICNet achieves

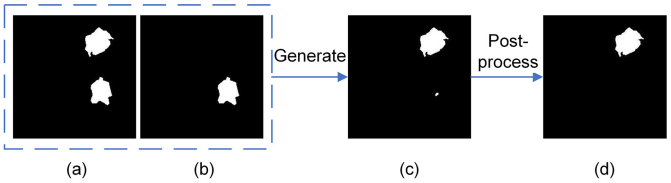


Fig. 5. Example of postprocessing to remove noise from the generated result of PICNet. (a) Original label L_O. (b) Mask M_L. (c) Preliminary generated result. (d) Postprocessed result L_G.

the overall disappearance of the target. This suggests that GSIA, which is designed to enable sub-instance-level changes, can also achieve instance-level changes.

Notably, PICNet belongs strictly to the category of image generation models rather than image editing models. However, in the label generation process of GSIA, the model is guided to expand (or contract) the boundary of the mining area according to our intentions by setting the mask outside (or inside) the boundary of the mining area. Thus, the entire label generation process can be regarded as an image editing process.

C. Image Generation

Yu et al. [26] proposed the DeepFill v2 model, based on their a previous study [47]. The DeepFill v2 model can perform image editing based on prior information from the labels. Therefore, DeepFill v2 was used as the basis of the proposed image editing model. As shown in Fig. 3, the DeepFill v2 model consists of two parts: the first generates coarse results and the second generates fine results that serve as the final results. This coarse-to-fine structure and the proposed gated convolution [47] improve the quality of the generated results.

Despite the high-quality images generated using DeepFill v2, the generated results did not match the guide labels well. As the complexity of the original image and guidance label increases, the model may generate results that seem more reasonable but do not match the label. To further enhance the guidance of the label information, a semantic segmentation module was introduced, as illustrated in Fig. 3. The DeepLab v3+ [48] model, pretrained on the mining area semantic data, was used as the semantic segmentation module to segment the generated result I_G, resulting in the predicted results (L_P). To exclude the interference outside the editing region M_I, the results of this area were replaced with the original label L_O, resulting in a new predicted result (L_{PM})

$$L_{\mathrm{PM}} = (1 - M_I) \times L_O + M_I \times L_P. \tag{1}$$

Subsequently, the semantic segmentation loss ($\mathrm{Loss}_{\mathrm{seg}}$) is calculated based on the similarity between L_{PM} and guiding label L_G

$$\mathrm{Loss}_{\mathrm{seg}} = |L_G - L_{\mathrm{PM}}|. \tag{2}$$

$\mathrm{Loss}_{\mathrm{seg}}$ optimizes the loss function by adjusting the generator loss and assisting in training DeepFill v2 as follows:

$$\mathrm{Loss} = \mathrm{Loss}_D + (1 - \lambda_{\mathrm{seg}}) \times \mathrm{Loss}_G + \lambda_{\mathrm{seg}} \times \mathrm{Loss}_{\mathrm{seg}} \tag{3}$$

where Loss_D denotes the total loss of the discriminator, Loss_G denotes the original total loss of the generator, and λ_{seg} denotes the optimization parameter of $\mathrm{Loss}_{\mathrm{seg}}$ to Loss_G.

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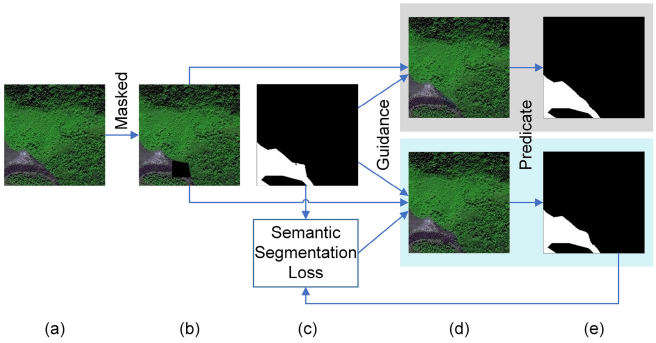


Fig. 6. Comparison of results with and without semantic segmentation loss Loss_{seg}. The top images in (d) and (e) show the results generated without using Loss_{seg}, whereas the bottom images use Loss_{seg}. (a) Original image I_O. (b) Masked original image by mask M_I. (c) Guiding label L_G. (d) Generated image I_G. (e) Predicted label L_{PM}.

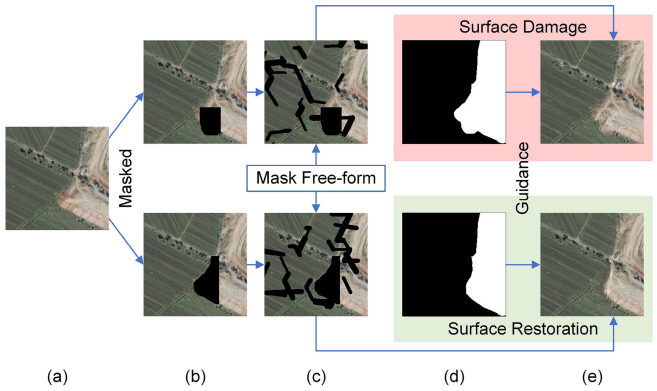


Fig. 7. Example of semantic image editing. The top images in (b)–(e) simulate the surface damage process during mining operations, whereas the bottom images simulate the surface restoration process. Notably, additional construction traces are simulated on the inner side of the mining area in (e). In addition, the top image in (e) simulates a gradual slope at the boundary of the mining area, whereas the bottom image simulates an abrupt slope. Furthermore, the mask Free-form does not damage the linear texture information of farmland and roads but only adds noise. (a) Original image I_O. (b) Masked original image by mask M_I. (c) Masked original image by mask M_I and Free-form. (d) Guiding label L_G. (e) Generated image I_G.

In this study, the value of λ_{seg} was gradually increased from 5% to 50% with the DeepFill v2 training, to strengthen the influence of the label L_G on the generated result I_G. In addition, using Loss_{seg} in the early model training stage enabled DeepFill v2 to learn the semantic information of the mining area and background.

With the assistance of Loss_{seg}, the model can generate results more similar to the guiding labels. As shown in Fig. 6, within the image editing region M_I, the guiding label L_G has a sharp protrusion relative to the original label L_O. Without semantic segmentation loss, the model would generate a mining area boundary based on a smooth curve, which was more similar to the original image. However, by incorporating Loss_{seg}, the model is forced to adhere to the guidance of the label L_G during image generation.

As shown in Fig. 3, different masks are used in the model training and data generation phases of DeepFill v2.

During the training phase of DeepFill v2, the mask used was also the random rectangular mask M_R, and the original label L_O was used to guide the model for training.

During the data generation stage, the characteristics of the mining area must be considered, as shown in Fig. 3. On the one hand, the real construction area in the mining area is often larger than the changed area, implying that construction traces exist within the mining area. On the other hand, the edges of mining areas exhibit gradual or abrupt slopes, which manifest as gradual or sudden changes in image textures. To account for these factors, the image mask M_I comprises the following three parts, as depicted in Fig. 7: 1) the previously generated change area label L_C, which is used to generate the corresponding image I_G for the label L_G; 2) the inner region of the mining area in M_R that corresponds to L_C and used to simulate construction traces other than the changed area L_C; and 3) a buffer area that is reserved to edit the nonmining area region to assist in simulating the edge features of the mining area. In addition, a mask Free-form [26] is employed to add noise and enhance the generalizability of the synthetic data.

D. Dataset Construction

Although each set of the original data is augmented to N sets (referred to as the augmentation multiplier N) after label and image generation, the amount of CD data (N_{CD}) obtained by random combination in the third step of Section III-A exceeds this number

$$N_{\text{CD}} = 2 \times N + N^2 \tag{4}$$

where $2 \times N$ represents the amount of CD data constructed using Strategy 1 in step 3 and N^2 represents the amount of CD data constructed using Strategy 2. Considering the large amount of data involved, a CD dataset was not created in advance. Instead, it was generated dynamically through random assignments during program runtime using the code. However, a higher augmentation multiplier N does not necessarily lead to better results, as detailed in Section V-A2.

IV. Experimental Results

A. Datasets

1) Open-Pit Mine CD Dataset: In this study, an open-pit mine CD dataset was created using multisource fusion RS images provided by a dynamic monitoring project of mine geological environmental changes in Liaoning, China. These images were acquired during 2018 and 2019, covering the period from June to December each year, and consisted of a multisource fusion of satellite data, including GF-1, GF-2, GF-6, Beijing-2, and ZY-3, with a spatial resolution ranging from 0.5 to 2 m. After preprocessing, such as image registration, these images were visually interpreted by experts to draw change maps by relevant standards [49], [50]. The identified areas of change primarily pertained to surface damage and restoration at the boundaries of the mining areas.

The images were resampled to a resolution of 1 m and cropped into nonoverlapping image patches of 256 × 256, resulting in 845 sets of open-pit mine CD datasets. The labels are binary maps, indicating whether have changes occurred. The CD dataset was randomly divided into training, validation, and test sets in a ratio of 6:2:2.

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2) Synthetic CD Dataset by GSIA: The bitemporal mining area semantic segmentation data used in GSIA were based on the same source images as those in Section IV-A1, and the feature labels were drawn based on the same identification principles. Subsequently, the images and labels were cropped into 256 × 256 patches, and the patches located at the boundary of the mining area were selected, guaranteeing that each patch contained both mining and nonmining areas. Finally, 8731 sets of bitemporal mine area data were obtained and used as the original data for the GSIA. The mining areas where the open-pit mine CD dataset was located were named Region 1, and the other areas were named Region 2. Region 1 showed changes in the mining area, whereas Region 2 did not. The purpose of this study was to investigate the effectiveness of GSIA using data from unchanged mining areas where real CD data are not available. Therefore, the original GSIA data were divided into two parts: 1830 sets for Region 1 and 6901 sets for Region 2.

In this study, the augmentation multiplier N was set to eight (as detailed in Section V-A2), resulting in each set of original data being synthesized into eight sets of open-pit mine CD data. Consequently, the data for Region 1 were augmented by GSIA to yield 14 640 sets of data; this augmentation method is denoted GSIA_1. Similarly, the data for Region 2 were augmented to produce 55 208 sets of data, named GSIA_2. The total amount of synthetic sets of data was 69 848. In addition, according to the definition of GSIA_1, only data related to the real CD training set were used for GSIA_1, whereas data related to the validation and test sets were excluded. Notably, as per (4), setting N to eight resulted in a final multiplier N_{CD} of 80 for the constructed CD data. Therefore, the total number of synthetic sets of open-pit CD data obtained was 698 480.

B. Implementation Details

The experiments were implemented using a desktop computer with an 11th Gen Intel i7-11700, 64-GB memory, and NVIDIA GeForce RTX3060, including evaluations of the synthetic image quality and data augmentation effectiveness of GSIA.

Image quality assessment was performed using three metrics: Inception Score (IS) [51], Fréchet inception distance (FID) [52], and kernel inception distance (KID) [53]. A higher IS value, a lower FID value, and a lower KID value indicate greater realism and diversity of the generated images. Moreover, because the KID value in this experiment was remarkably low, it was multiplied by 1000 (KID × 1000) for evaluation purposes.

To evaluate the effectiveness of data augmentation, five advanced supervised deep-learning CD models were used as baselines: FC-Siam-conc [54], SNUNet [55], BIT [56], A2Net [57], and DMINet [58]. The mean values of the five models were used as comprehensive evaluation results. During baseline training, the batch size was set to eight, the initial learning rate was 0.001, the optimizer was Adam, and 200 epochs were trained. Moreover, given the limited availability of real open-pit CD datasets, fourfold cross validation was used to improve the reliability of the detection results.

The data augmentation methods used for comparison with GSIA included nine basic methods and five advanced methods. In addition, the combination of the nine basic methods was used for comparison.

1) *Flip:* This includes horizontal and vertical flipping.
2) *Rotate:* Random rotation. To eliminate the black edges generated by rotation, an inner square was obtained and resampled to the original image size.
3) *Crop:* This includes randomly cropping any rectangle on the image and then resampling it to the original image size.
4) *Color:* This includes changes in brightness, contrast, and saturation.
5) *Overlap:* This refers to an RS image-cropping method that maintains a certain degree of overlap compared to nonoverlapping methods, resulting in a larger number of datasets. The overlap rate was set to 80% in this study.
6) *Noise:* Gaussian noise.
7) *Mosaic [19]:* Four images are used to randomly crop and compose a new image.
8) *Cutout [17]:* This includes randomly erasing rectangles on the image.
9) *CutMix [18]:* The rectangular region of one image is replaced with the same-sized rectangular region of another image.
10) *Basic:* The aforementioned nine data augmentation methods are common. The data augmentation method that simultaneously uses these nine methods is referred to as the Basic method.
11) *YOCO [31]:* This involves cutting one image into two pieces and performing various data augmentations individually within each piece.
12) *RandomMix [32]:* This incorporates Mixup [59], CutMix, ResizeMix [60], and Fmix [61] to enhance robustness. However, Mixup was not applicable to this experiment and was therefore excluded.
13) *ObjectAug [20]:* This involves decoupling the image into instances and the background and performing various data augmentations on each instance individually.
14) *BS [21]:* This refers to swapping bitemporal backgrounds to eliminate interference from background information.
15) *CopyPaste [22]:* This involves copying instance patches and pasting them into areas without instances in RS images.

Among the methods mentioned above, Flip, Rotate, Crop, Color, Overlap, and Noise belong to image-level methods; Mosaic, Cutout, CutMix, YOCO, and RandomMix belong to region-level methods; and ObjectAug, BS, and CopyPaste belong to instance-level methods. Moreover, in this experiment, image translation was not included in the comparison because of the generation of black borders caused by image shifting, whereas image overlapping was considered a superior alternative. Furthermore, equal opportunities for each method were ensured by setting the probability of using each method to 50%.

To quantitatively evaluate the effectiveness of the data augmentation methods quantitatively, four evaluation metrics were

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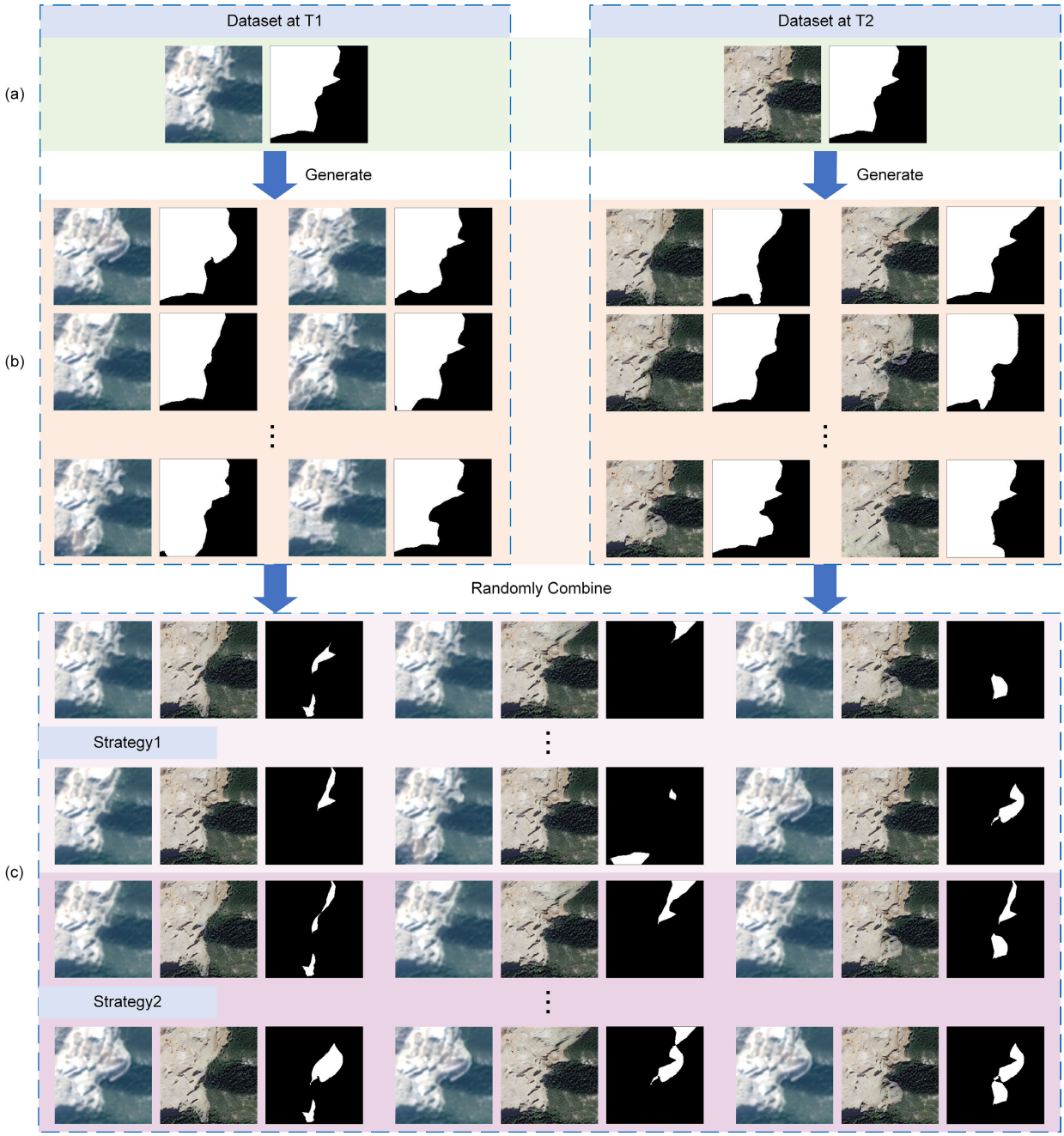


Fig. 8. Example of GSIA synthesizing multiple sets of CD data for open-pit mines based on bitemporal semantic segmentation data within the mining areas. (a) Original data I_O and L_O. (b) Generated data I_G and L_G. (c) Open-pit mine CD dataset synthesized based on two strategies.

calculated: recall (Rec), precision (Pre), overall accuracy (OA), and $F1$-score ($F1$). Higher values of these metrics indicate better detection results. Notably, the $F1$-score simultaneously considers precision and recall, rendering it more suitable for detecting imbalanced samples in the foreground and background and thus serving as the most reliable evaluation metric

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{7}$$

$$F1 = \frac{2}{\text{Rec}^{-1} + \text{Pre}^{-1}} \tag{8}$$

where TP, TN, FP, and FN represent the numbers of true positives, true negatives, false positives, and false negatives, respectively.

C. Evaluation of Synthetic Image Quality

Fig. 8 shows the powerful data generation capability of GSIA, confirming that the GSIA method meets the requirements of both authenticity and diversity in synthesizing the open-pit mine CD dataset. We can observe that the shadows are effectively simulated.

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TABLE I
EVALUATION OF SYNTHETIC IMAGE QUALITY OF GSIA

Mask		IS↑	FID↓	KID*1000↓
M_I	Free-form			
√		5.01	2.04	0.75
√	√	**5.04**	**2.03**	**0.73**

Based on the generated data from $GSIA_2$, the generated image quality of using only mask M_I and using both mask M_I and Free-form was evaluated. The results in Table I show low FID and KID values for both cases, indicating effective data generation by the GSIA. In addition, the IS might not be sufficiently high because the synthetic data are not generated based on the ImageNet dataset [51]. Furthermore, the addition of the Free-form resulted in an increase in IS and a decrease in both FID and KID, suggesting that the Free-form improved the generalization ability of the model and enhanced the diversity of the generated images.

D. Evaluation of Date Augmentation Effectiveness

1) Method Comparison: Fourteen data augmentation methods were used for the experimental comparison with GSIA in the open-pit mine CD, with the original CD training set sizes of 100%, as listed in Table II. The results demonstrated the following: 1) GSIA achieved the best performance, and specifically, $GSIA_1 + GSIA_2$ produced the best results, with mean OA values of 98.12% and mean $F1$ values of 83.67%; 2) both $GSIA_1$ and $GSIA_2$ outperformed the other methods, indicating that using $GSIA_1$ alone without additional mining areas or $GSIA_2$ only with other unchanged mining areas can significantly improve the CD model recognition performance, and moreover, combining $GSIA_1$ and $GSIA_2$ further enhanced the augmentation effect; however, the benefits may decrease because of the redundancy in the GSIA dataset, as detailed in Section V-A4; and 3) in the case of imbalanced foreground and background samples, the model tended to prioritize improving the precision over recall. This is evident from the significantly higher precision of the baseline compared to the recall. Therefore, in this experiment, a greater demand exists for the improvement of the model in recall. GSIA involves synthesizing foreground targets (i.e., change areas in CD) to allocate more attention to the foreground, leading to improvements in recall rates, and finally achieving better recognition performance. By contrast, most other methods render it difficult for the CD model to concentrate on the foreground, resulting in a relatively lower $F1$ score. It is worth noting that CopyPaste, as the only method outside of GSIA capable of synthesizing change areas in CD, also improves recall. However, this method operates at the instance level rather than the sub-instance level, leading to a decrease in precision. Additional analysis is detailed in Section V-B1.

The question of whether synthetic CD data generated from other data sources can truly replace real data has not been investigated [13], [22], [23], [24], which is a critical indicator of the reliability of the synthesized data. Moreover, when the original open-pit mine CD training set was restricted to 0%, most data augmentation methods could not be applied. Therefore, only the effectiveness of CopyPaste and $GSIA_2$ were verified, as shown in Table III. A comparison of the results in Tables II and III reveals the following.

1) The performance of using only $GSIA_2$ is significantly better than using only CopyPaste, and it reaches the level of using all real data without data augmentation. Specifically, using only $GSIA_2$ leads to mean OA and $F1$ values of 97.64% and 78.65%, respectively, which are much higher than the 80.94% and 35.07% obtained with using only CopyPaste and slightly higher than 97.57% and 77.27% obtained with using all real data.
2) $GSIA_2$ exhibits a notably higher recall than precision while maintaining an acceptable level of precision. In contrast, CopyPaste exhibits a high recall but low precision. These are consistent with analysis 3 in Table II.

To illustrate the effect of the reduction in data quantity on different data augmentation methods, the increase in the $F1$ value of the methods concerning the baseline was explored for the limited training sets of 100%, 60%, and 20%, as shown in Fig. 9. The scenario in which the training set was 0% was not analyzed because only $GSIA_2$ and CopyPaste were effective. The results suggested that in open-pit mine CD, the effectiveness of different data augmentation methods exhibits varying trends as the amount of available data decreases. Among the other methods, better methods, such as Flip and Rotate, are largely unaffected by the training set quantity, whereas poorer methods, such as Noise and CutMix, may result in weaker augmentation effects or even reduced the model detection capability as the data quantity decreases, as detailed in Section V-B1. $GSIA_1 + GSIA_2$ improved the mean $F1$ value from 6.40% to 6.85% to 12.56% as the dataset size decreased from 100% to 60% to 20%, respectively. This is because GSIA is a data augmentation method that does not rely on the original open-pit CD data. Instead, its augmentation effect was solely related to the amount of data in the synthetic dataset generated by GSIA, independent of the real CD dataset quantity. $GSIA_1$ is a special case, as its definition implies that the amount of synthetic data decreases as the size of the real training set decreases. Therefore, this method does not result in a significant increase in the augmentation effect as the size of the original dataset decreases.

2) Compatibility Study: To investigate the compatibility between GSIA and basic augmentation methods and to compare GSIA with other advanced methods, experiments were conducted with a 100% training set, as listed in Table IV. Comparing Tables II and IV reveals the following.

1) The Basic method for the fusion of multiple data augmentation methods can achieve better results. However, comparing the results of Basic in Table IV with those of $GSIA_1 + GSIA_2$ in Table IV with the 100% training set shows that GSIA outperformed the fusion of multiple other methods.
2) GSIA does not conflict with basic methods and can be combined to achieve better results. Even after applying Basic, GSIA improved the effectiveness of model detection. Specifically, $GSIA_1$, $GSIA_2$, and $GSIA_1 + GSIA_2$

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TABLE II
COMPARISON OF DATA AUGMENTATION METHODS

Method		Rec / Pre / OA / F1 (%)					
(Baseline+)		FC-Siam-conc	SNUNet	BIT	A2Net	DMINet	Mean
No Augmentation		68.41 / 80.62 / 97.26 / 74.01	73.51 / 81.92 / 97.54 / 77.49	77.51 / 78.86 / 97.53 / 78.18	66.12 / 89.76 / 97.67 / 76.15	78.65 / 82.48 / 97.84 / 80.52	72.84 / 82.73 / 97.57 / 77.27
Image Level	Flip	71.07 / 85.03 / 97.63 / 77.43	79.49 / 82.55 / 97.85 / 80.99	79.01 / 81.06 / 97.72 / 80.02	71.32 / 90.91 / 97.97 / 79.93	83.20 / 82.52 / 98.01 / 82.86	76.82 / 84.41 / 97.84 / 80.25
	Rotate	71.19 / **85.67** / 97.64 / 77.76	80.60 / 83.63 / 97.98 / 82.09	80.46 / 81.73 / 97.76 / 81.09	72.59 / 90.78 / 98.00 / 80.67	81.45 / 83.25 / 97.97 / 82.34	77.26 / 85.01 / 97.87 / 80.79
	Crop	68.96 / 84.03 / 97.47 / 75.76	75.78 / 82.67 / 97.61 / 79.07	78.75 / 80.25 / 97.66 / 79.49	66.15 / 92.94 / 97.69 / 77.29	80.90 / 82.10 / 97.88 / 81.50	74.11 / 84.40 / 97.66 / 78.62
	Color	66.58 / 83.96 / 97.38 / 74.27	72.85 / 82.34 / 97.57 / 77.31	76.66 / 81.28 / 97.72 / 78.91	64.98 / 93.69 / 97.75 / 76.74	75.71 / 86.86 / 97.93 / 80.90	71.36 / 85.63 / 97.67 / 77.62
	Overlap	69.26 / 80.64 / 97.29 / 74.52	76.45 / 81.04 / 97.63 / 78.68	78.09 / 80.43 / 97.60 / 79.24	65.24 / 93.95 / 97.74 / 77.01	81.73 / 81.05 / 97.97 / 81.39	74.16 / 83.42 / 97.65 / 78.17
	Noise	68.02 / 81.18 / 97.23 / 74.02	72.77 / 81.39 / 97.51 / 76.84	76.06 / 80.58 / 97.61 / 78.25	65.91 / 90.50 / 97.66 / 76.27	80.63 / 79.94 / 97.55 / 80.28	72.68 / 82.72 / 97.51 / 77.13
Region Level	Mosaic	68.77 / 80.44 / 97.21 / 74.15	73.74 / 81.53 / 97.44 / 77.44	76.90 / 79.84 / 97.47 / 78.34	63.87 / 93.83 / 97.64 / 76.00	80.07 / 82.37 / 97.88 / 81.20	72.67 / 83.60 / 97.53 / 77.43
	Cutout	67.40 / 82.16 / 97.26 / 74.05	73.71 / 82.02 / 97.58 / 77.64	77.09 / 80.89 / 97.66 / 78.95	65.86 / 92.75 / 97.70 / 77.02	77.65 / 83.20 / 97.82 / 80.33	72.34 / 84.21 / 97.60 / 77.60
	CutMix	67.90 / 81.86 / 97.28 / 74.23	74.47 / 81.25 / 97.50 / 77.71	77.97 / 78.12 / 97.58 / 78.04	64.40 / 93.20 / 97.64 / 76.17	77.63 / 83.10 / 97.80 / 80.27	72.47 / 83.51 / 97.56 / 77.29
	YOCO	70.01 / 85.93 / 97.59 / 77.16	76.78 / 84.73 / 97.86 / 80.56	77.23 / 82.58 / 97.81 / 79.81	66.60 / **93.99** / 97.78 / 77.96	77.71 / **87.44** / 97.96 / 82.28	73.67 / **86.93** / 97.80 / 79.56
	RandomMix	68.31 / 82.40 / 97.33 / 74.70	74.77 / 80.63 / 97.41 / 77.59	77.81 / 77.98 / 97.41 / 77.90	64.46 / 90.96 / 97.54 / 75.46	78.15 / 82.52 / 97.76 / 80.28	72.70 / 82.90 / 97.49 / 77.18
Instance Level	ObjectAug	66.24 / 86.27 / 97.43 / 74.93	71.46 / **85.42** / 97.64 / 77.82	73.81 / **83.35** / 97.61 / 78.29	66.33 / 89.69 / 97.61 / 76.26	77.60 / 83.54 / 97.82 / 80.46	71.09 / 85.65 / 97.62 / 77.55
	BS	67.16 / 83.89 / 97.45 / 74.60	73.73 / 82.89 / 97.49 / 78.04	78.27 / 79.62 / 97.61 / 78.94	70.20 / 84.85 / 97.60 / 76.83	76.68 / 85.85 / 97.92 / 81.01	73.21 / 83.42 / 97.61 / 77.88
	CopyPaste	**81.60** / 67.73 / 96.97 / 74.02	83.71 / 72.67 / 97.36 / 77.80	**87.85** / 70.34 / 97.27 / 78.13	75.21 / 78.34 / 97.48 / 76.75	89.51 / 73.40 / 97.64 / 80.66	83.58 / 72.50 / 97.35 / 77.47
Sub-instance Level	$GSIA_1$	78.50 / 83.07 / 97.77 / 80.72	**85.30** / 81.84 / 98.07 / 83.53	81.95 / 82.63 / 98.01 / 82.29	84.87 / 82.48 / 98.13 / 83.66	88.80 / 80.11 / 98.12 / 84.23	83.88 / 82.03 / 98.02 / 82.89
	$GSIA_2$	81.39 / 79.41 / 97.78 / 80.39	82.09 / 83.26 / 98.01 / 82.67	82.95 / 79.99 / 97.85 / 81.44	82.40 / 86.79 / 98.24 / 84.54	91.43 / 76.27 / 98.03 / 83.17	84.05 / 81.14 / 97.98 / 82.44
	$GSIA_1$+$GSIA_2$	81.31 / 81.95 / **97.87** / **81.63**	83.95 / 84.13 / **98.18** / **84.04**	84.38 / 81.14 / **98.06** / **82.73**	**86.99** / 82.94 / **98.26** / **84.92**	**93.31** / 78.15 / **98.24** / **85.06**	**85.99** / 81.66 / **98.12** / **83.67**

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TABLE III
EVALUATION OF $GSIA_2$ AND COPYPASTE FOR TRAINING SET SIZE OF 0%

Method		(%)	FC-Siam-conc	SNUNet	BIT	A2Net	DMINet	Mean
Instance Level	CopyPaste	Rec	**84.05**	**91.92**	**83.35**	61.72	79.18	80.04
		Pre	16.07	15.06	20.27	33.05	33.24	23.54
		OA	73.77	68.80	81.33	91.01	89.77	80.94
		F1	26.98	25.89	32.61	43.05	46.83	35.07
Sub-Instance Level	$GSIA_2$	Rec	75.56	85.90	80.69	**81.55**	**85.25**	**81.79**
		Pre	**73.34**	**75.55**	**76.41**	**77.10**	**76.56**	**75.79**
		OA	**97.29**	**97.71**	**97.51**	**97.88**	**97.79**	**97.64**
		F1	**74.43**	**80.39**	**78.49**	**79.26**	**80.67**	**78.65**

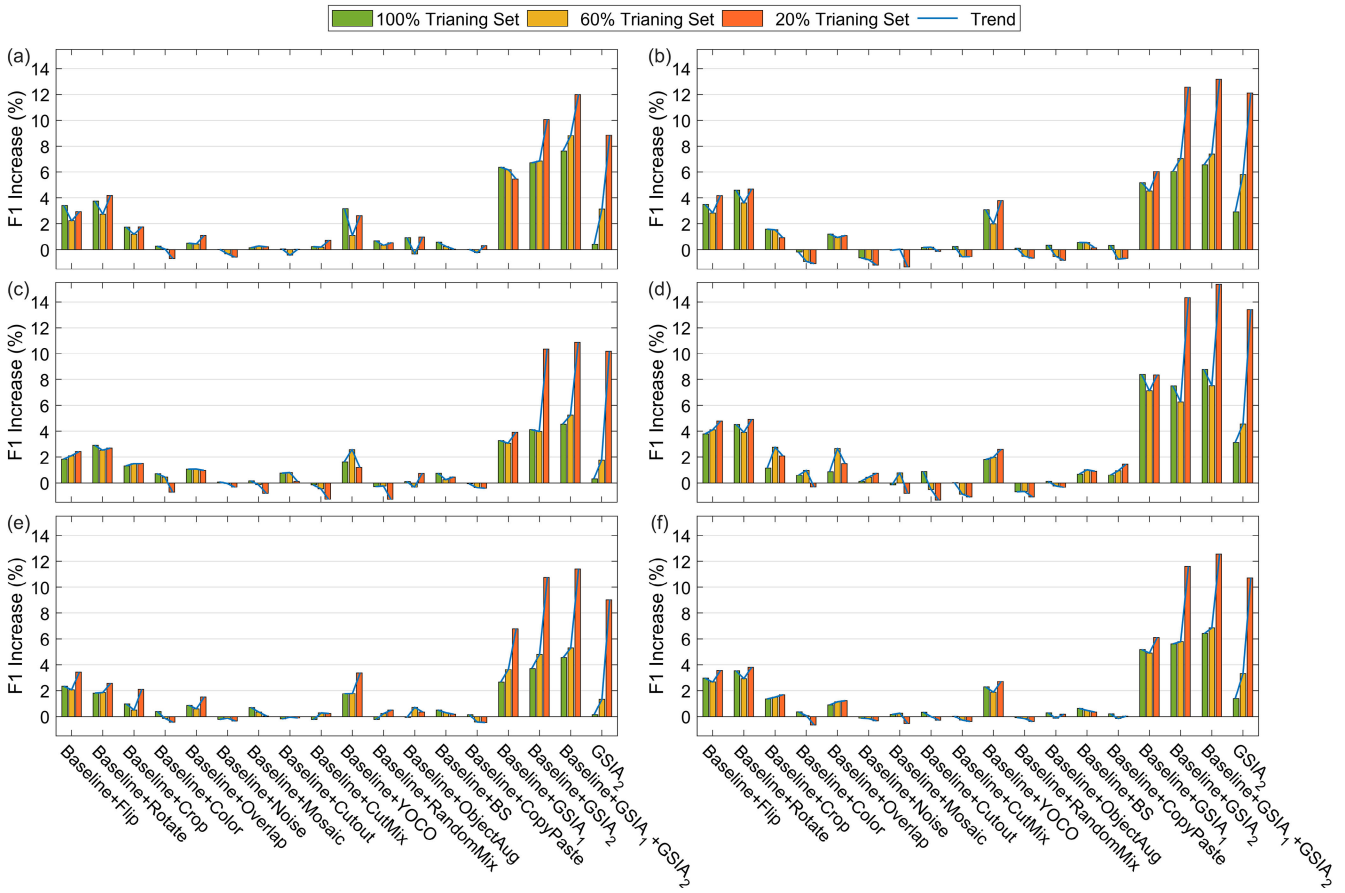


Fig. 9. Comparison of the effect of reduction in data quantity on different data augmentation methods. (a) FC-Siam-conc. (b) SNUNet. (c) BIT. (d) A2Net. (e) DMINet. (f) Mean.

showed mean OA improvements of 0.28%, 0.32%, and 0.38%, reaching 98.25%, 98.29%, and 98.35%, respectively. Their mean $F1$ values were enhanced by 2.99%, 3.22%, and 3.77%, reaching 84.70%, 84.93%, and 85.47%, respectively.

3) In contrast, other advanced methods do not show significant improvement when combined with the Basic method. For example, while YOCO exhibits a noticeable enhancement over the Baseline, it does not yield substantial additional gains when applied on top of the Basic, possibly because YOCO is a subset of Basic. Furthermore, although CopyPaste improves significantly on recall and complements Basic, differences in application scenarios result in unsatisfactory precision performance, consistent with analysis 3 in Table II.

Fig. 10 visualizes the partial results obtained using the SNUNet-CD model in the compatibility experiments. As shown in the figure, GSIA significantly improves the recognition accuracy of CD. Fig. 10 visualizes the partial results obtained using the SNUNet model in the compatibility experiments. As shown in the figure, GSIA significantly improved the recognition accuracy of CD.

V. DISCUSSION

A. Parameter Analysis

To explore the influence of certain parameters on the performance of the GSIA method, the training set was restricted to 0%, and the experiments were performed solely on $GSIA_2$, thereby eliminating other potential confounding factors.

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TABLE IV
EVALUATION OF COMPATIBILITY BETWEEN ADVANCED AUGMENTATION METHODS AND BASIC AUGMENTATION METHODS

Method (Baseline+)		(%)	FC-Siam-conc	SNUNet	BIT	A2Net	DMINet	Mean
No Augmentation		Rec	68.41	73.51	77.51	66.12	78.65	72.84
		Pre	80.62	81.92	78.86	89.76	82.48	82.73
		OA	97.26	97.54	97.53	97.67	97.84	97.57
		F1	74.01	77.49	78.18	76.15	80.52	77.27
Image Level+ Region Level	Basic	Rec	73.23	83.90	82.77	72.04	84.41	79.27
		Pre	84.69	82.45	80.56	93.07	83.48	84.85
		OA	97.70	98.02	98.00	98.02	98.11	97.97
		F1	78.54	83.17	81.65	81.21	83.95	81.70
	Basic+YOCO	Rec	72.76	84.27	83.73	70.87	84.27	79.18
		Pre	85.50	82.47	80.54	**94.46**	84.39	85.47
		OA	97.71	98.06	97.96	98.01	98.20	97.99
		F1	78.62	83.36	82.10	80.98	84.33	81.88
	Basic+RandomMix	Rec	72.81	83.42	83.90	72.29	85.89	79.66
		Pre	84.72	81.68	78.54	92.41	83.80	84.23
		OA	97.60	97.97	97.87	97.83	98.29	97.91
		F1	78.31	82.54	81.13	81.12	84.83	81.59
Image Level+ Region Level+ Instance Level	Basic+ObjectAug	Rec	71.70	84.06	83.68	71.28	83.87	78.92
		Pre	85.78	82.39	80.23	93.69	85.04	85.43
		OA	97.72	98.06	97.91	97.99	98.22	97.98
		F1	78.11	83.22	81.92	80.96	84.45	81.73
	Basic+BS	Rec	74.04	83.89	84.66	72.37	84.27	79.62
		Pre	83.19	83.37	79.24	93.23	84.39	84.77
		OA	97.57	98.07	97.98	98.03	98.20	97.96
		F1	78.35	83.63	81.86	81.48	84.33	81.86
	Basic+CopyPaste	Rec	74.71	**91.91**	**91.17**	82.04	88.64	85.69
		Pre	83.20	75.36	72.74	81.62	79.85	78.56
		OA	97.61	97.94	97.75	98.00	98.08	97.88
		F1	78.73	82.82	80.92	81.83	84.02	81.66
Image Level+ Region Level+ Sub-Instance Level	Basic+$GSIA_1$	Rec	79.32	83.79	83.37	83.30	88.47	83.65
		Pre	84.99	**87.30**	**85.84**	87.69	83.44	**85.85**
		OA	97.96	98.28	98.35	98.33	98.34	98.25
		F1	82.06	85.51	84.59	85.44	85.88	84.70
	Basic+$GSIA_2$	Rec	78.27	85.86	84.85	85.93	87.48	84.48
		Pre	85.28	86.05	84.79	85.17	**85.90**	85.44
		OA	97.95	**98.43**	98.30	98.33	98.45	98.29
		F1	81.63	85.96	84.82	85.55	86.68	84.93
	Basic+$GSIA_1$+$GSIA_2$	Rec	**79.78**	86.15	86.17	**87.80**	**88.95**	**85.77**
		Pre	**85.54**	86.22	84.50	84.45	85.52	85.25
		OA	**98.04**	98.40	**98.37**	**98.36**	**98.58**	**98.35**
		F1	**82.56**	**86.19**	**85.32**	**86.09**	**87.20**	**85.47**

1) Combination Strategy: As discussed in Section III-D, two combination strategies were employed during dataset construction to further expand the synthetic open-pit mine CD dataset. As listed in Table V, both strategies proved beneficial for improving the final results. However, the effectiveness of these two strategies varies across different CD models, which, along with the augmentation multiplier N, is expected to affect the final results, as detailed in Section V-A2.

2) Augmentation Multiplier N: According to (4), with the assistance of Strategy 2, the amount of synthetic open-pit mine CD data N_{CD} can increase exponentially with an increase in the augmentation multiplier N. To investigate whether the amplification effect of N on the CD model can reach the theoretical value, N was set to start from one and doubled until 16, as shown in Fig. 11. We can observe that the optimal performance is achieved when $N = 8$. Although increasing N is expected to improve the detection capability of the CD models by increasing N_{CD}, the experimental results showed an initial increase, followed by a slight decline in the $F1$ value. This may be owing to the following reasons: 1) information redundancy exists between the CD data generated by Strategies 1 and 2 and 2) the number of random masks that can be generated for an image is limited, and the corresponding randomly generated labels and images may yield similar results, leading to data redundancy. Although the $F1$ value may decrease slightly after reaching a certain value as N increases, the impact is small. Moreover, the augmentation data generated by GSIA were automatically generated without manual labor. Therefore, researchers can consider limiting N according to specific research requirements.

In addition, we note from (4) that Strategy 2 exerts a dominant influence as the augmentation multiplier N increases.

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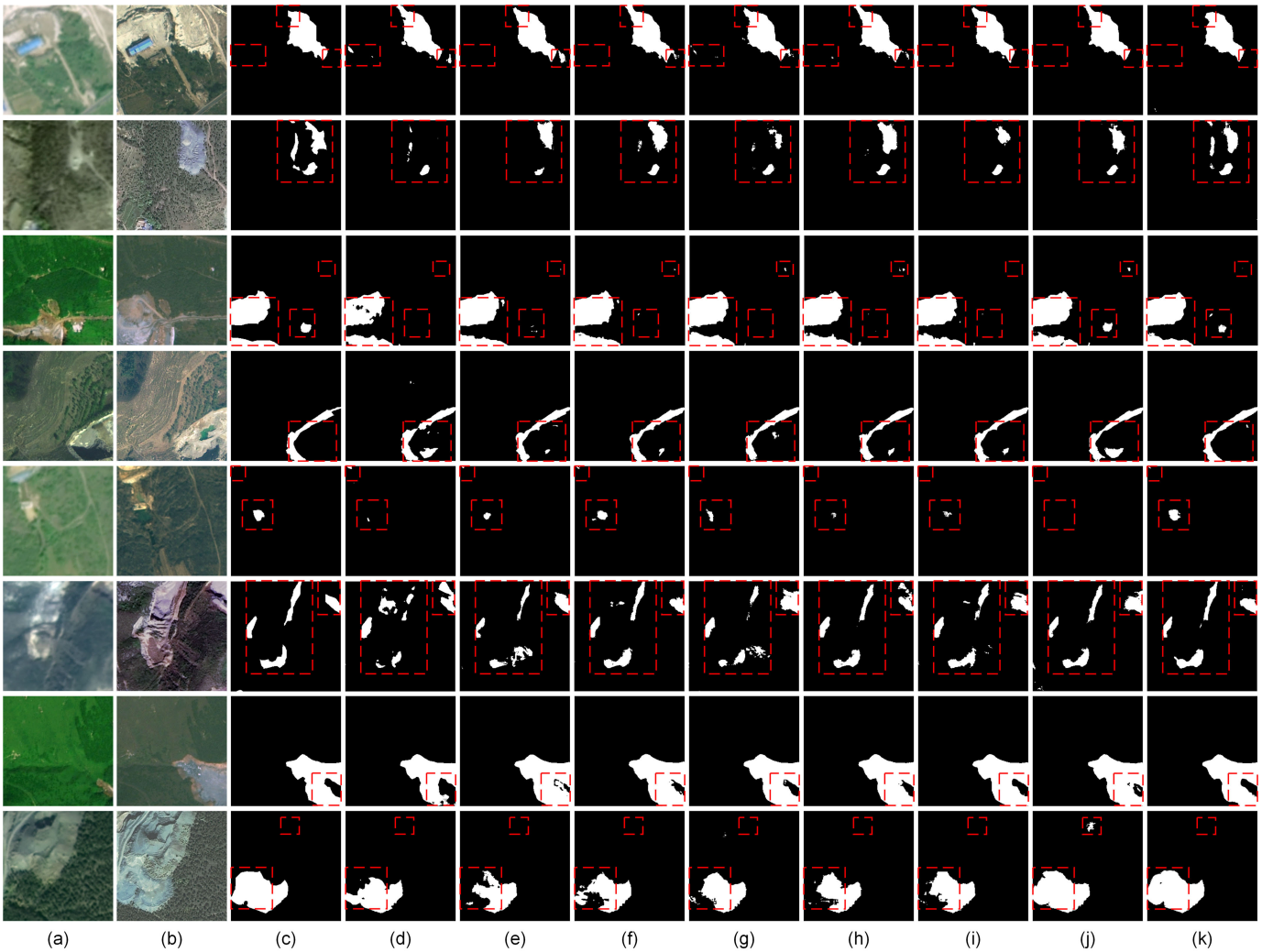


Fig. 10. Comparison of SNUNet-CD results using different methods in compatibility experiments. The red boxes in (c)–(k) highlight the differences in detection results among the methods. (a) Image at $T1$. (b) Image at $T2$. (c) Ground truth. (d) Baseline. (e) Baseline + Basic. (f) Baseline + Basic + YOCO. (g) Baseline + Basic + RandomMix. (h) Baseline + Basic + ObjectAug. (i) Baseline + Basic + BS. (j) Baseline + Basic + CopyPaste. (k) Baseline + Basic + $GSIA_1$ + $GSIA_2$.

TABLE V
EVALUATION OF THE TWO COMBINATION STRATEGIES

Strategy		(%)	FC-Siam-conc	SNUNet	BIT	A2Net	DMINet	Mean
$Strategy_1$	$Strategy_2$							
√		Rec	68.18	82.04	79.48	80.62	81.62	78.39
		Pre	75.10	74.01	74.36	75.61	**78.06**	75.43
		OA	97.02	97.50	97.36	97.77	97.67	97.46
		F1	71.47	77.82	76.84	78.04	79.80	76.79
	√	Rec	70.75	83.16	**83.06**	79.05	84.72	80.15
		Pre	**78.45**	74.52	71.11	76.53	75.87	75.29
		OA	**97.30**	97.38	97.29	97.72	97.69	97.48
		F1	74.40	78.61	76.62	77.77	80.05	77.49
√	√	Rec	**75.56**	**85.90**	80.69	**81.55**	**85.25**	**81.79**
		Pre	73.34	**75.55**	**76.41**	**77.10**	76.56	**75.79**
		OA	97.29	**97.71**	**97.51**	**97.88**	**97.79**	**97.64**
		F1	**74.43**	**80.39**	**78.49**	**79.26**	**80.67**	**78.65**

A comprehensive analysis of Table V and Fig. 11 reveals that when Strategy 2 outperforms Strategy 1, such as in FC-Siam-conc, SNUNet, and DMINet, the $F1$ value of the CD model increases rapidly at the initial stage of increasing N. Conversely, when Strategy 2 efficacy is relatively poor, such as in BIT and A2Net, the $F1$ value quickly

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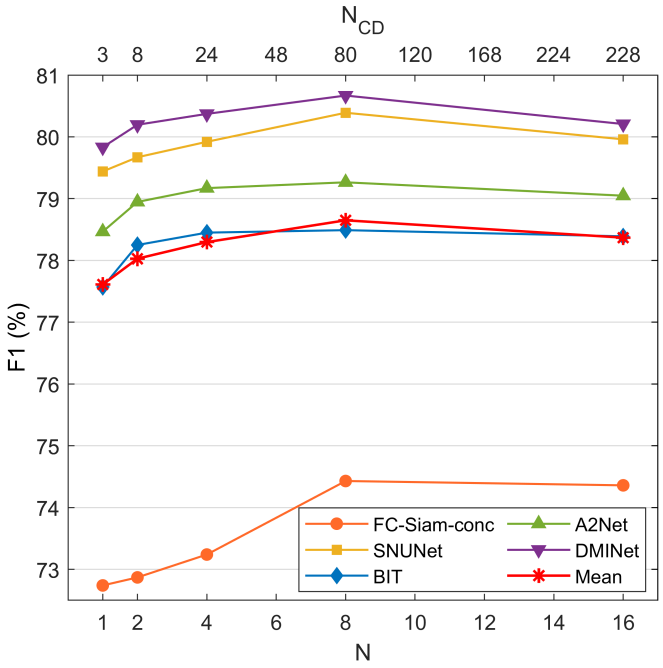


Fig. 11. Analysis of the effect of augmentation multiplier N on the amount of synthetic CD data N_{CD} and the accuracy of the CD model.

reaches a bottleneck, and setting N to two or four may be considered.

3) Rectangle Mask M_R Area: The areas of the rectangular mask M_R, image editing region, and change region L were positively correlated. The recognition accuracy of the CD model could be improved by increasing the proportion of L in the image. However, expanding the image editing area can reduce the quality of the generated images. Therefore, the impact of M_R on the final performance of data augmentation in open-pit mine CD must be analyzed.

Synthetic data generated with an augmentation multiplier N of 16 were stratified into 16 levels based on the size of the M_R areas. The image generation quality and CD recognition accuracy were then computed for each size level in a process referred to as size-based stratification. By contrast, the control group was obtained by randomly selecting data of any size with an equivalent number of samples, which is referred to as random selection (corresponding to the synthetic data generated under $N = 1$). The experimental results are presented in Fig. 12. Fig. 12(a) reveals a strong correlation between IS, FID, and KID values with the M_R area, indicating that the larger the M_R area, the lower the image quality. Fig. 12(b) shows that the $F1$ score first increases rapidly and then gradually with the increase in M_R area.

Compared with the optimal selection of M_R areas by size-based stratification, random selection strategy may be more practical and effective. The differences in the detection of object features among studies can lead to inconsistencies in optimal selections by size-based stratification, whereas random selection may yield near-optimal results. As illustrated in Fig. 12, the M_R area in random selection is similar in size to Level 10 in size-based stratification, but with a higher image quality and $F1$ score. This improved performance can be attributed to the increased generalization ability of data with different area sizes. Furthermore, the performance of open-pit mine CD in random selection was similar to the optimal result in size-based stratification. Thus, extensive experimental analysis is not required to determine the appropriate M_R area when applying the model to other studies. Alternatively, it may be more appropriate to consider M_R with varying area sizes.

4) Quantity of Synthetic CD Data: In general, deep-learning models achieve higher performance with larger amounts of data; however, the benefits gradually diminish. Given that the source data for GSIA are collected manually and require substantial effort, investigating the impact of data volume on the final performance of data augmentation is important to reduce labor costs. To this end, experiments were conducted using synthetic open-pit mine CD datasets with sizes of 25%, 50%, 75%, and 100%. The results depicted in Fig. 13 indicate that CD recognition accuracy improves with increasing synthetic data volume. However, the extent of improvement gradually diminishes. Thus, in practical applications, the manual workload can be appropriately reduced to achieve a certain level of recognition performance based on the demand.

B. Domain Adaptation Study

FID was used to measure the domain metric distance (or correlation) between the training and test sets in each experiment. Although commonly used to evaluate GAN generation, its fundamental principle is to assess the similarity of distributions between two datasets, making it useful for measuring domain distances. In addition, solely evaluating image correlations is insufficient for CD tasks. Bitemporal image difference maps are more reflective of the CD data characteristics. Therefore, both the images and difference maps were subjected to correlation analysis, and the mean values were used as the final results.

1) Method Comparison for Data Augmentation: In Section IV-D1, we observed that different augmentation methods exhibit varying performances, and certain methods may even result in a decrease in CD recognition effectiveness. This can be attributed to two factors. First, the relatively limited size of the original dataset enables GSIA, as a data augmentation method independent of the original dataset, to achieve the best performance in the experiments. This is evident from Fig. 9, where a decrease in the data size results in a weakening of the $F1$ value for certain augmentation methods, such as Noise and CutMix. Second, the varying principles among the different methods might not necessarily be suitable for this experiment, as elucidated in Table II and Fig. 9.

To further validate the impact of the second factor, a domain distance analysis was conducted between the training data generated by various data augmentation methods and the test set. Moreover, to ensure a balance between the methods, the generated data for each method were set to four times the original data volume. The results of the domain distance analysis are listed in Table VI. Noise, Mosaic, Cutout, CutMix, RandomMix, and CopyPaste exhibited larger domain distances, whereas Overlap and GSIA exhibited smaller domain distances, which is consistent with the results shown in Table II and Fig. 9. This suggests that the suboptimal performance of methods, such as noise, may be attributed to the significant

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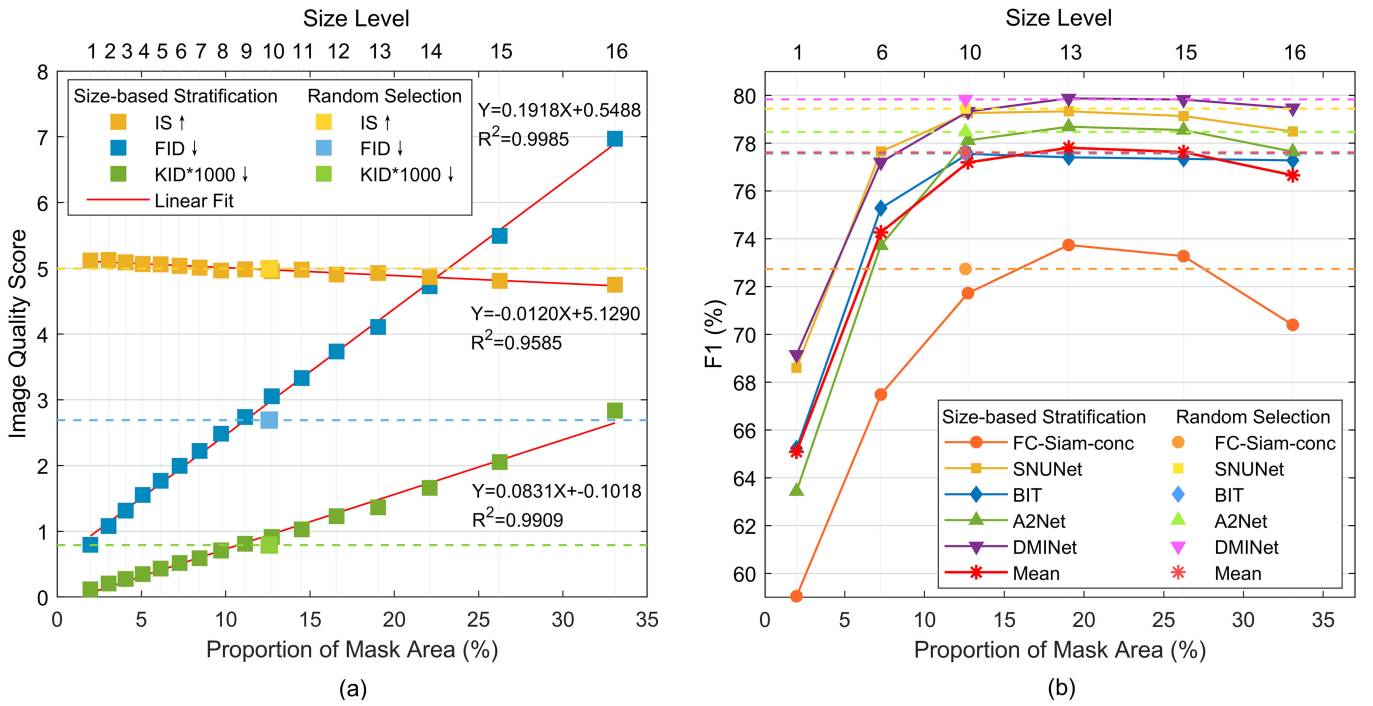


Fig. 12. Analysis of the effect of the rectangular mask M_R area on GSIA. (a) Effect of M_R area on synthetic image quality. (b) Effect of M_R area on the open-pit mine CD accuracy.

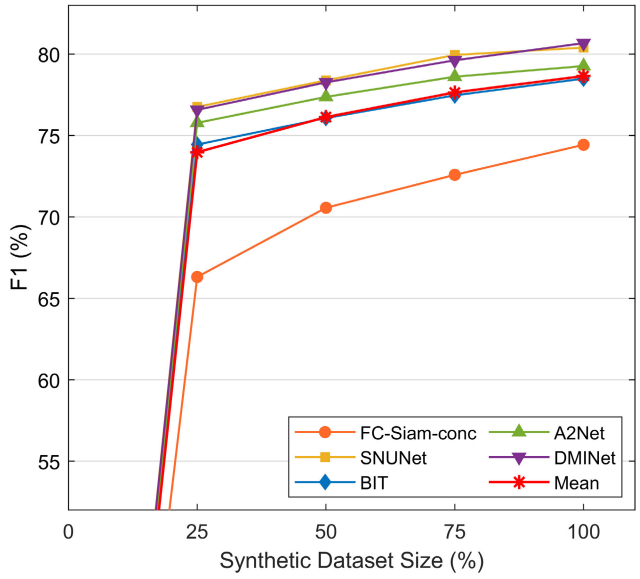


Fig. 13. Analysis of the effect of synthetic open-pit mine CD dataset size on the accuracy of the CD model.

dissimilarity between the augmented data and original data distributions resulting from these transformations. However, the situation with Color was different. Although the domain distance is not significantly large, the corresponding $F1$ values are unsatisfactory. This could be attributed to the greater emphasis on enhancing precision, which made a limited contribution to $F1$ in this experiment, as shown in Table II.

2) Domain Adaptation Study of GSIA: When transferring a pretrained model to a new area, factors, such as the region, sensor, lighting, and season, can cause a decrease in the CD model recognition performance. GSIA utilizes data from other mining areas for data augmentation. Therefore, this approach is applicable not only for data augmentation but also for domain adaptation based on synthetic CD datasets from the target study area.

TABLE VI
DOMAIN ADAPTATION STUDY FOR DATA AUGMENTATION METHODS

Method		FID		
		Image	Difference Map	Mean
Baseline		88.22	129.17	108.7
Image Level	Flip	103.92	147.55	125.74
	Rotate	114.59	155.55	135.07
	Crop	101.52	148.51	125.01
	Color	104.43	148.46	126.45
	Overlap	76.29	111.02	93.65
	Noise	167.94	185.22	176.58
Region Level	Mosaic	160.58	193.65	177.11
	Cutout	155.58	195.68	175.63
	CutMix	130.63	161.48	146.06
	YOCO	109.60	148.33	128.96
	RandomMix	143.94	166.60	155.27
Instance Level	ObjectAug	120.60	155.41	138.00
	BS	111.65	154.00	132.82
	CopyPaste	131.24	169.21	150.22
Sub-instance Level	$GSIA_1$	94.35	125.26	109.81
	$GSIA_2$	80.06	113.77	96.91
	$GSIA_1$+$GSIA_2$	86.21	116.92	101.56

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TABLE VII
COMPARISON OF DATA AUGMENTATION METHODS FOR DOMAIN ADAPTATION

Method ($Baseline_A$+)	(%)	FC-Siam-conc	SNUNet	BIT	A2Net	DMINet	Mean
No Augmentation	Rec	65.52	72.16	76.86	66.33	78.11	71.80
	Pre	80.84	80.23	76.94	**84.33**	80.09	80.49
	OA	96.49	96.46	96.95	96.94	97.10	96.79
	F1	72.38	75.98	76.90	74.26	79.09	75.72
$Basic_A$	Rec	65.74	74.31	82.44	68.72	83.22	74.89
	Pre	79.95	79.68	74.66	83.92	77.39	79.12
	OA	96.51	96.91	96.85	96.95	97.13	96.87
	F1	72.15	76.90	78.36	75.56	80.20	76.63
$Basic_A$+$GSIA_A$	Rec	66.74	77.24	75.47	78.44	82.10	76.00
	Pre	79.45	76.19	80.56	80.38	78.00	78.91
	OA	96.53	96.85	97.10	97.65	97.22	97.07
	F1	72.54	76.71	77.93	79.40	80.00	77.32
$Basic_A$+$GSIA_A$	Rec	**85.05**	90.94	**88.50**	88.40	90.68	**88.71**
	Pre	78.81	80.09	80.41	84.04	**83.85**	81.44
	OA	97.41	97.81	97.74	98.10	98.15	97.84
	F1	81.81	85.17	84.26	86.17	87.13	84.91
$Basic_A$+$GSIA_A$+$GSIA_B$	Rec	84.96	**90.97**	87.13	**88.62**	**91.48**	88.63
	Pre	**82.94**	**82.13**	**82.42**	84.27	83.73	**83.10**
	OA	**97.75**	**98.02**	**97.79**	**98.21**	**98.19**	**97.99**
	F1	**83.94**	**86.32**	**84.71**	**86.39**	**87.43**	**85.76**

TABLE VIII
DOMAIN ADAPTATION STUDY OF GSIA. THE VARIOUS SCENARIOS ARE RANKED IN DESCENDING ORDER OF THEORETICAL SIMILARITY BETWEEN THE DATASETS. "80%" ETC. IN THE TABLE INDICATES THE PROPORTION OF A CERTAIN BASIC DOMAIN. "-" INDICATES NO RELEVANT EXPERIMENTS IN THIS ARTICLE. DOMAIN ANALYSIS OF "-" IS ONLY CONDUCTED TO LINK RANDOM PARTITIONING AND DOMAIN-BASED PARTITIONING EXPERIMENTS AND PREVENT THE INTERFERENCE OF TEST SET VARIABILITY ON FID ANALYSIS

Partitioning Method	Related Experiment	Domain		FID		
		Training Set	Test Set	Image	Difference Map	Mean
Random	Baseline	80%A+20%B	80%A+20%B	**88.22**	129.17	**108.70**
	$GSIA_1$+$GSIA_2$	80%$GSIA_A$+20%$GSIA_B$	80%A+20%B	98.26	**128.26**	113.26
	-	50%$GSIA_A$+50%$GSIA_B$	80%A+20%B	108.10	129.40	118.75
Domain-based	$GSIA_B$	$GSIA_B$	B	111.52	140.99	126.26
	$GSIA_A$+$GSIA_B$	50%$GSIA_A$+50%$GSIA_B$	B	109.19	136.76	122.98
	-	80%$GSIA_A$+20%$GSIA_B$	B	120.00	140.14	130.07
	$GSIA_B$	$GSIA_A$	B	122.65	152.73	137.69
	$Baseline_A$	A	B	117.90	158.53	138.22

In the data partitioning for the domain adaptation experiments, the open-pit mine CD dataset was partitioned based on the differences in the regions, sensors, lighting, seasons, and other factors that constitute the differences in domains. Specifically, the dataset was divided into two domains with a 4:1 ratio, where four parts were utilized for training and validation (referred to as source domain A), and one part was used for testing (referred to as target domain B). The training and validation datasets were then merged and randomly partitioned into training and validation sets at a 3:1 ratio. This approach of partitioning data based on a domain is called domain-based partitioning, whereas the previous method of dividing data without considering the domain is called random partitioning.

Based on this, experiments similar to those described in Section IV-D2 were conducted, as shown in Table VII. Subscripts A and B denote domains A and B, respectively. By comparing Tables IV and VII, the following can be inferred.

1) Compared to the results of random partitioning, the mean recall, precision, OA, and $F1$ values of $Baseline_A$ using domain-based partitioning decreased.
2) Although $Basic_A$ still improved the CD model identification capabilities, its performance was significantly lower with domain-based partitioning than with random partitioning, with the mean increase in the OA and $F1$ values dropping from 0.40% to 0.08% and from 4.43% to 0.91%, respectively.
3) $GSIA_A$ did not show any significant enhancement effect, with increases of only 0.20% and 0.69% in the OA and $F1$ values, respectively. This is likely owing to the differences between the source domains A and B,

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which may be further amplified by $GSIA_A$. Conversely, $GSIA_B$ significantly improved the CD model identification capabilities, with increases of 0.97% and 8.28% in the OA and $F1$ values, respectively. Both results demonstrated the effectiveness of applying GSIA to domain adaptation.

4) Based on the results of "$Baseline_A$ + $Basic_A$ + $GSIA_A$ + $GSIA_B$" and the analysis in 3), we note that utilizing GSIA synthetic data from source domain A in combination with data from target domain B can enhance the CD model generalization ability, leading to improved detection performance.

To further validate the previous analysis, the domain distances between the training and test sets in the experiments presented in Tables IV and VII were analyzed, as shown in Table VIII. To facilitate analysis, validation sets were incorporated into the training sets. The experiment "$Baseline_A$ + $Basic_A$ + $GSIA_A$ + $GSIA_B$" in Table VII is substituted with the experiment "$GSIA_A$ + $GSIA_B$" (similarly for other experiments). During the domain analysis, four basic domains were defined: A, B, $GSIA_A$, and $GSIA_B$. The domains of the various experimental datasets can be represented by combining them in certain proportions. In addition, the data volume for each experiment was set to be consistent with the original data volume.

The FID results in Table VIII validate the results presented in Table VII. The mean FID values are generally arranged in ascending order, which is consistent with the theoretical analysis. Notably, the FID values for the experimental "$GSIA_A$ + $GSIA_B$" are lower than those for "$GSIA_B$." Although this may appear counterintuitive, it is in agreement with the previous analysis 4) for Table VII that under the support of target domain data, source domain data can improve the generalization ability. These findings suggest that the problem of an insufficient correlation between the training and test sets in domain adaptation can be effectively addressed using GSIA.

VI. Conclusion

GSIA was proposed to address the issue of data scarcity in supervised deep-learning-based open-pit mine CD. GSIA improves GAN-based local image editing to construct sub-instance-level transformations with finer scales than instance-level transformations. Based on this, GSIA achieves label and image synthesis using bitemporal semantic segmentation data of unchanged mining areas. Furthermore, the synthetic open-pit CD dataset can be expanded further by random combination strategies of GSIA. Both qualitative and quantitative evaluations demonstrate the authenticity and diversity of the synthetic CD dataset based on GSIA. To verify the effectiveness of GSIA for data augmentation, 14 data augmentation methods were compared based on five CD models, with GSIA exhibiting the best performance. GSIA is complementary to basic augmentation methods and can further improve the accuracy of CD recognition. GSIA is highly robust and does not synthesize biased samples, even when the original CD dataset size is reduced. Notably, when the original training data were gradually reduced to zero, training the CD models using only the GSIA-synthesized data achieved the same performance as training on 100% of the original data. In addition to data augmentation, GSIA performed well in domain adaptation. CD experimental results and domain distance analysis demonstrated an improvement in GSIA in the correlation between the training and test sets.

The GSIA method, although designed for mining areas, can be extended to other CD tasks involving sub-instance-level transformations, such as construction sites, vegetation, and rivers. However, modifications to GSIA may be necessary for targets with instance-level changes, such as buildings, pedestrians, and cars. Moreover, extending instance-level transformations to sub-instance-level transformations is crucial. For example, although most changes in buildings are instance-level, phenomena, such as house extensions or demolitions, also involve sub-instance-level changes. In addition, GSIA can be improved by focusing on label and image editing models to further enhance the diversity and authenticity of the synthetic data. Future work may also include automatic or semiautomatic generation of semantic segmentation labels to reduce the cost of manual mapping.

References

[1] W. Chen, X. Li, H. He, and L. Wang, "A review of fine-scale land use and land cover classification in open-pit mining areas by remote sensing techniques," *Remote Sens.*, vol. 10, no. 2, p. 15, Dec. 2017, doi: 10.3390/rs10010015.

[2] Pratiwi et al., "Managing and reforesting degraded post-mining landscape in Indonesia: A review," *Land*, vol. 10, no. 6, p. 658, Jun. 2021, doi: 10.3390/land10060658.

[3] Z. Yang, Y. Shen, J. Li, H. Jiang, and L. Zhao, "Unsupervised monitoring of vegetation in a surface coal mining region based on NDVI time series," *Environ. Sci. Pollut. Res.*, vol. 29, no. 18, pp. 26539–26548, Apr. 2022, doi: 10.1007/s11356-021-17696-9.

[4] X. Nie, Z. Hu, M. Ruan, Q. Zhu, and H. Sun, "Remote-sensing evaluation and temporal and spatial change detection of ecological environment quality in coal-mining areas," *Remote Sens.*, vol. 14, no. 2, p. 345, Jan. 2022, doi: 10.3390/rs14020345.

[5] Y. Han et al., "Tracking vegetation degradation and recovery in multiple mining areas in Beijing, China, based on time-series Landsat imagery," *GISci. Remote Sens.*, vol. 58, no. 8, pp. 1477–1496, Nov. 2021, doi: 10.1080/15481603.2021.1996319.

[6] A. Barenblitt et al., "The large footprint of small-scale artisanal gold mining in Ghana," *Sci. Total Environ.*, vol. 781, Aug. 2021, Art. no. 146644, doi: 10.1016/j.scitotenv.2021.146644.

[7] Z. Lv, T. Liu, J. A. Benediktsson, and N. Falco, "Land cover change detection techniques: Very-high-resolution optical images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 44–63, Mar. 2022, doi: 10.1109/MGRS.2021.3088865.

[8] S. Du, W. Li, J. Li, S. Du, C. Zhang, and Y. Sun, "Open-pit mine change detection from high resolution remote sensing images using DA-UNet++ and object-based approach," *Int. J. Mining, Reclamation Environ.*, vol. 36, no. 7, pp. 512–535, Aug. 2022, doi: 10.1080/17480930.2022.2072102.

[9] C. Tang et al., "An improved fully convolution network model for change detection in mining areas using Sentinel-2 images," *Remote Sens. Lett.*, vol. 12, no. 7, pp. 684–694, Jul. 2021, doi: 10.1080/2150704X.2021.1925372.

[10] S. Camalan et al., "Change detection of Amazonian alluvial gold mining using deep learning and Sentinel-2 imagery," *Remote Sens.*, vol. 14, no. 7, p. 1746, Apr. 2022, doi: 10.3390/rs14071746.

[11] J. Gallwey, C. Robiati, J. Coggan, D. Vogt, and M. Eyre, "A Sentinel-2 based multispectral convolutional neural network for detecting artisanal small-scale mining in ghana: Applying deep learning to shallow mining," *Remote Sens. Environ.*, vol. 248, Oct. 2020, Art. no. 111970, doi: 10.1016/j.rse.2020.111970.

[12] S. Kuny, H. Hammer, and A. Thiele, "CNN based vehicle track detection in coherent SAR imagery: An analysis of data augmentation," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2022, pp. 93–98, May 2022, doi: 10.5194/isprs-archives-XLIII-B1-2022-93-2022.

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[13] X. Luo et al., "Research on change detection method of high-resolution remote sensing images based on subpixel convolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1447–1457, 2021, doi: 10.1109/JSTARS.2020.3044060.

[14] R. Raghavan, D. C. Verma, D. Pandey, R. Anand, B. K. Pandey, and H. Singh, "Optimized building extraction from high-resolution satellite imagery using deep learning," *Multimedia Tools Appl.*, vol. 81, no. 29, pp. 42309–42323, Dec. 2022, doi: 10.1007/s11042-022-13493-9.

[15] F. Cui and J. Jiang, "Shuffle-CDNet: A lightweight network for change detection of bitemporal remote-sensing images," *Remote Sens.*, vol. 14, no. 15, p. 3548, Jul. 2022, doi: 10.3390/rs14153548.

[16] J. Ding, B. Chen, H. Liu, and M. Huang, "Convolutional neural network with data augmentation for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 364–368, Mar. 2016, doi: 10.1109/LGRS.2015.2513754.

[17] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," *arXiv:1708.04552*, 2017.

[18] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, Oct. 2019, pp. 6022–6031.

[19] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[20] J. Zhang, Y. Zhang, and X. Xu, "ObjectAug: Object-level data augmentation for semantic image segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Shenzhen, China, Jul. 2021, pp. 1–8.

[21] H. Chen, W. Li, S. Chen, and Z. Shi, "Semantic-aware dense representation learning for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630018, doi: 10.1109/TGRS.2022.3203769.

[22] L. Yan, J. Yang, and Y. Zhang, "Building instance change detection from high spatial resolution remote sensing images using improved instance segmentation architecture," *J. Indian Soc. Remote Sens.*, vol. 50, no. 12, pp. 2317–2336, Dec. 2022, doi: 10.1007/s12524-022-01601-z.

[23] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603216, doi: 10.1109/TGRS.2021.3066802.

[24] X. Rui, Y. Cao, X. Yuan, Y. Kang, and W. Song, "DisasterGAN: Generative adversarial networks for remote sensing disaster image generation," *Remote Sens.*, vol. 13, no. 21, p. 4284, Oct. 2021, doi: 10.3390/rs13214284.

[25] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019, doi: 10.1109/TGRS.2018.2858817.

[26] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. Huang, "Free-form image inpainting with gated convolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, Oct. 2019, pp. 4470–4479.

[27] Y. Zeng, Z. Lin, and V. M. Patel, "SketchEdit: Mask-free local image manipulation with partial sketches," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 5941–5951.

[28] C. Chen et al., "Remote sensing image augmentation based on text description for waterside change detection," *Remote Sens.*, vol. 13, no. 10, p. 1894, May 2021, doi: 10.3390/rs13101894.

[29] Y. Cao, B. Sui, and W. Zhang, "REL-SAGAN: Relative generation adversarial network integrated with attention mechanism for scene data augmentation of remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3107–3119, 2022, doi: 10.1109/JSTARS.2022.3166927.

[30] Y. Zhang et al., "Diversifying tire-defect image generation based on generative adversarial network," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022, doi: 10.1109/TIM.2022.3160542.

[31] J. Han et al., "You only cut once: Boosting data augmentation with a single cut," 2022, *arXiv:2201.12078*.

[32] X. Liu, F. Shen, J. Zhao, and C. Nie, "RandomMix: A mixed sample data augmentation method with multiple mixed modes," 2022, *arXiv:2205.08728*.

[33] Y. Aaren, G. Bebis, and M. Nicolescu, "Two-step data augmentation for masked face detection and recognition: Turning fake masks to real," in *Proc. 2nd Int. Conf. Image Process. Vis. Eng.*, 2022, pp. 126–134.

[34] L. Wang, W. Chen, W. Yang, F. Bi, and F. R. Yu, "A state-of-the-art review on image synthesis with generative adversarial networks," *IEEE Access*, vol. 8, pp. 63514–63537, 2020, doi: 10.1109/ACCESS.2020.2982224.

[35] E. Pajouheshgar, T. Zhang, and S. Süsstrunk, "Optimizing latent space directions for GAN-based local image editing," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 1740–1744.

[36] D. Wu et al., "Random occlusion recovery for person re-identification," *J. Imag. Sci. Technol.*, vol. 63, no. 3, pp. 030405-1–030405-9, May 2019, doi: 10.2352/J.ImagingSci.Technol.2019.63.3.030405.

[37] L. Zhang, C. Long, Q. Yan, X. Zhang, and C. Xiao, "CLA-GAN: A context and lightness aware generative adversarial network for shadow removal," *Comput. Graph. Forum*, vol. 39, no. 7, pp. 483–494, Oct. 2020, doi: 10.1111/cgf.14161.

[38] T. Nagae, R. Abiko, T. Yamaguchi, and M. Ikehara, "Shadow detection and removal using GAN," in *Proc. 28th Eur. Signal Process. Conf. (EUSIPCO)*, Amsterdam, The Netherlands, Jan. 2021, pp. 630–634.

[39] S. Gu, J. Bao, H. Yang, D. Chen, F. Wen, and L. Yuan, "Mask-guided portrait editing with conditional GANs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 3431–3440.

[40] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 5967–5976.

[41] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021, doi: 10.1016/j.isprsjprs.2021.07.007.

[42] A. Bulat, J. Yang, and G. Tzimiropoulos, "To learn image super-resolution, use a GAN to learn how to do image degradation first," in *Proc. 15th Eur. Conf. Comput. Vis.*, Munich, Germany, 2018, pp. 187–202.

[43] G. Antipov, M. Baccouche, and J.-L. Dugelay, "Face aging with conditional generative adversarial networks," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Beijing, China, Sep. 2017, pp. 2089–2093.

[44] Y. Shen, C. Yang, X. Tang, and B. Zhou, "InterFaceGAN: Interpreting the disentangled face representation learned by GANs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2004–2018, Apr. 2022, doi: 10.1109/TPAMI.2020.3034267.

[45] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 4396–4405.

[46] C. Zheng, T.-J. Cham, and J. Cai, "Pluralistic image completion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 1438–1447.

[47] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, "Generative image inpainting with contextual attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 5505–5514.

[48] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, vol. 2018, pp. 833–851.

[49] *Specification for Investigation and Evaluation of Mine Geological Environment, DD2014-05*, China Geological Surv., Beijing, China, 2014.

[50] *Technical Requirements for Remote Sensing Monitoring of Mineral Resource Development, DD2011-06*, China Geological Surv., Beijing, China, 2011.

[51] S. Barratt and R. Sharma, "A note on the inception score," 2018, *arXiv:1801.01973*.

[52] M. Hensel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. 31st Annu. Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 1–12.

[53] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, "Demystifying MMD GANs," 2018, *arXiv:1801.01401*.

[54] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 4063–4067.

[55] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3056416.

[56] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

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[57] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.
[58] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.
[59] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.
[60] J. Qin, J. Fang, Q. Zhang, W. Liu, X. Wang, and X. Wang, "ResizeMix: Mixing data with preserved object information and true labels," 2020, *arXiv:2012.11101*.
[61] E. Harris, A. Marcu, M. Painter, M. Niranjan, A. Prügel-Bennett, and J. Hare, "FMix: Enhancing mixed sample data augmentation," 2020, *arXiv:2002.12047*.



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